



University
Mohammed VI
Polytechnic



Deliverable #: Title of Lab/Project here (e.g.: Conceptual Design)

Data Management Course
UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	UNO
Member 1	Daniel Tchouang
Member 2	Amine Sira
Member 3	Mohammed Oumane
Member 4	Zaid Zehari
Repository Link	https://github.com/unocomputing/unodatabase.git

1 Roles

- Daniel: Report and SQL code
- Sira: Report
- Mohammed: SQL code
- Zaid; SQL code

2 Challenges

No significant challenges were encountered during the project work

3 Justification of conversion choices

For **Person**, all attributes are required (**NOT NULL**) except **end_date**, which is optional.

For **Student** and **Employee**, since they are children of **Person**, we gave them a primary key that is also a foreign key to **Person**. We did the same for every other child in the hierarchy (**Academic**, **NonAcademic**, **Faculty**, **Administrative**, **Technical**), so each one's primary key comes from its direct parent, all the way back to **Person**.

For **Maintenance** and **CalibrationRecord**, since they are weak entities owned by **EquipmentUnit**, their primary key is the owner's key plus a partial key (the start times-tamp / calibration date).

For relationships where an entity must participate at least once (e.g. every equipment unit must belong to a model, every reservation must be made by someone), we made the foreign key **NOT NULL** to enforce this.

We can't guarantee that a project has exactly one **PI**: nothing stops two people from both being recorded with **role = PI** on the same project, since a unique constraint on **role** would force it to be unique across everyone, not just within one project.

We also can't guarantee some "at least one" relationships, e.g. that every laboratory is funded by at least one budget, or that every laboratory has at least one attached person — these depend on rows existing in another table, which a foreign key alone can't force.

4 Assumptions

- Due to lack of examples for this system, we assume that most primary keys like code, id, etc. are integers, except serial number, model id, and budget line, which we made varchar.
- We assume attributes representing money are floating values, since we don't know the currency.
- We suppose we don't need hours, minutes, and seconds for time attributes, so they are all of type date.
- We give the default value "Pending" to the status attribute of **Reservation**, because we assume that for reservations needing approval, the decision might not be made right away.